

Today's Content

- ① Negative numbers Representation
- ② Left Shift & Right Shift
- ③ 5 Interesting / Simple Bitwise problems
- ④ 1 Advanced Bitwise problem

8bit

⑩ : 0 0 0 0 1 0 1 0

-10 : 1 0 0 0 1 0 1 0 $\times \Rightarrow -128 + 8 + 2 \Rightarrow -118$

2's complement = $-a = \sim a + 1$ } Proof in Today's Doubts

~ 10 : 1 1 1 1 0 1 0 1

+1 : 0 0 0 0 0 0 0 1

1 1 1 1 0 1 1 0 $\xrightarrow{2^7/2}$ Decimal Representation

-128 64 32 16 8 4 2 1

$\Rightarrow -128 + 64 + 32 + 16 + 4 + 2 = -128 + 118 = -10$

$\sim a$ is Nothing but
its Complement

$$23 : \quad \underline{0} \quad \underline{0} \quad \underline{0} \quad \underline{1} \quad \underline{0} \quad \underline{1} \quad \underline{1} \quad \underline{1}$$

$$\sim 23 : \quad \begin{array}{cccccccc} 1 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \end{array} \quad \left. \vphantom{\begin{array}{cccccccc} 1 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \end{array}} \right\}$$

$$+1 \quad \begin{array}{cccccccc} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{array}$$

$$\begin{array}{cccccccc} 1 & 1 & 1 & 0 & 1 & 0 & 0 & 1 \\ \downarrow & \downarrow & \downarrow & & \downarrow & & & \downarrow \\ -128 & +64 & +32 & & +8 & & & +1 \end{array}$$

$$-128 + 105 = -23$$

$$30 : \quad \underline{0} \quad \underline{0} \quad \underline{0} \quad \underline{1} \quad \underline{1} \quad \underline{1} \quad \underline{1} \quad \underline{0}$$

$$\sim 30 : \quad \begin{array}{ccccccc} 1 & 1 & 1 & 0 & 0 & 0 & 0 \end{array} \quad \left. \vphantom{\begin{array}{ccccccc} 1 & 1 & 1 & 0 & 0 & 0 & 0 \end{array}} \right\} \begin{array}{c} 1 \\ - \end{array}$$

$$+1 : \quad \begin{array}{ccccccc} 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \quad \left. \vphantom{\begin{array}{ccccccc} 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}} \right\} \begin{array}{c} 1 \\ - \end{array}$$

$$\begin{array}{ccccccc} 1 & 1 & 1 & 0 & 0 & 0 & 1 \\ \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ -128 & +64 & +32 & & & & +2 \end{array}$$

$$-128 + 98 = -30$$

Left Shift

8 bit Number

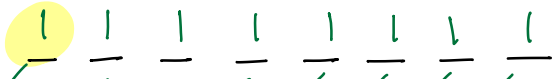
		7	6	5	4	3	2	1	0	
		-2	2	2	2	2	2	2	2	
$a = 10$:	0	0	0	0	1	0	1	0		: 10
$a \ll 1$:	0	0	0	1	0	1	0	0		: $2^4 + 2^2 = 20$: 10×2
$a \ll 2$:	0	0	1	0	1	0	0	0		: $2^5 + 2^3 = 40$: 10×2^2
$a \ll 3$:	0	1	0	1	0	0	0	0		: $2^6 + 2^4 = 80$: 10×2^3
$a \ll 4$:	1	0	1	0	0	0	0	0		: $-2^7 + 2^5 = -128 + 32 = -96$

$a \times 2^4 \Rightarrow 10 \times 2^4 \Rightarrow 160$
 $a = 10$
 In 8 bit Number : $[-128, 127]$

overflow & bit cannot store 160

$a = 127 =$ <table border="0"> <tr><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr> <tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td></tr> <tr><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr> </table> -128	0	1	1	1	1	1	1	1	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	$a = 15, 15 \ll 2$ <table border="0"> <tr><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td></tr> <tr><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr> <tr><td></td><td></td><td>32</td><td>16</td><td>8</td><td>4</td><td></td><td></td></tr> </table> $\Rightarrow 60$ $\Rightarrow 15 \times 2^2$	0	0	0	0	1	1	1	1	0	0	1	1	1	1	0	0			32	16	8	4		
0	1	1	1	1	1	1	1																																										
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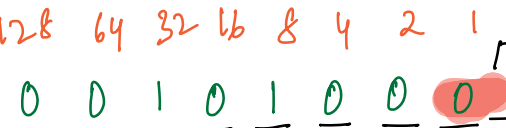
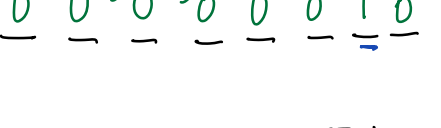
$a \ll n = a \times 2^n$

$a = -1$:  Decimal :
 $a \ll 1$:  = -2

$-1 \ll 1 = -1 * 2^1 = -2 \Rightarrow$ 8 bit Number $\rightarrow$ No Overflow

Right Shift

8 bit Number

$a = 40$:  LSB discarded $\rightarrow$ Dec $\rightarrow 40$:
 $a \gg 1$:  : $2^4 + 2^2 \Rightarrow 20 = 40/2$
 $a \gg 2$:  : $2^3 + 2^1 \Rightarrow 10 = 40/2^2$
 $a \gg 3$:  : $40/2^3 \Rightarrow 5$
 $a \gg 4$:  : 40/16 $\Rightarrow 2$
Integer division

$a = 50$: $a \gg 2 : 50/2^2 \Rightarrow 50/4 \Rightarrow 12$

$a = 10$: $a \gg 4 : 10/2^4 \Rightarrow 10/16 \Rightarrow 0$

+ve positive Numbers

$a \gg N : a/2^N$

No overflow

TODO : Right Shift for Negative Numbers

Problems

→ Check if i^{th} bit of N is Set or Unset

$\{1 \leq N \leq 10^9\}$, $[0 \leq i \leq 30]$ → i is bit position

bool CheckBit (int N , i) {

$N = 21$, $i = 2 \rightarrow$ Set

↳ $\begin{array}{cccccc} & 4 & 3 & 2 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & \end{array}$

$N = 25$, $i = 2 \rightarrow$ unset

↳ $\begin{array}{cccccc} & 4 & 3 & 2 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 & \end{array}$

Example

$N = 106$: $\begin{array}{cccccc} 7 & 6 & 5 & 4 & 3 & 2 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 1 & 0 \end{array}$

$i = 0$: $(N \& 1) == 1$, set else unset

$N = 106$: $\begin{array}{cccccc} 7 & 6 & 5 & 4 & 3 & 2 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 1 & 0 \end{array}$

$i = 1$: $(N \gg 1) \& 1 == 1$

$N = 24$, $i = 2 \rightarrow$ unset → false

↳ $\begin{array}{cccccc} & 4 & 3 & 2 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & \end{array}$

$N = 14$, $i = 3$

↳ $\begin{array}{cccc} & 3 & 2 & 1 & 0 \\ & 1 & 1 & 1 & 0 \end{array}$

↳ Set → True

$N = 106$

$\begin{array}{cccccc} 7 & 6 & 5 & 4 & 3 & 2 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 1 & 0 \end{array}$

$i = 0$: $(N \& 1) \neq 0$

$\begin{array}{cccccc} 7 & 6 & 5 & 4 & 3 & 2 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{array}$

$i = 2$

$N \& (1 \ll 2) == 0$

2nd bit is unset

$N = 106$: 01101010
 7 6 5 4 3 2 1 0
 (Bit 3 is highlighted)

$i = 3$: $(N \gg i) \& 1 == 1$

```

bool checkBit (N, i) {
  if ((N >> i) & 1 == 1) {
    return True;
  }
  else {
    return False;
  }
  return (N >> i) & 1 == 1;
}
  
```

```

bool checkBit (N, i) {
  if (N & (1 << i) != 0) {
    return true;
  }
  else {
    return false;
  }
}
  
```

LO: 30 PM hw

Note: All bitwise operators only on Integer Datatypes

$8 = 00001000$
 7 6 5 4 3 2 1 0
 (Bit 3 is highlighted)

$i = 3$ (1 << 3)

$N \& (1 << 3) != 0$

3rd bit is Set

Examples

$a = 10$

$a \& 10 \rightarrow 20$

print(a) $\rightarrow 10$

$a = 10$

$a \gg 1 \rightarrow 5$

print(a) $\rightarrow 10$

Given Integer (N) calculate No: of Set bits are present

Count Set Bits (N) : $0 \leq N \leq 10^9$

Int : 32 bits

$N = 25 : 11001 : 3 \text{ bits}$
 $\uparrow \uparrow \quad \uparrow$

$N = 40 : 101000 : 2 \text{ bits}$

$N = 35 : 100011 : 3 \text{ bits}$

[31, 30... 0]
 0

countSet Bits (N) {

cnt = 0

i = 0; i < 32; i++ {

if (checkBit(N, i)) {

cnt++

}

return cnt;

if $N = 27 : 11011$

$N = 27$, 1st code: 31 iterations

$N = 109$

6 5 4 3 2 1 0
 1 1 0 1 1 0 1

$N > 1$:

while (N > 0) {

(N & 1) == 1

// cnt++

N = N > 1

N = N / 2

return cnt; $\approx (\log_2 N)$

$N = 27$, 2nd code: 5 iterations

→ Prerequisites

$$\begin{array}{r}
 \begin{array}{cccc}
 & 4 & 3 & 2 & 1 & 0 \\
 11 & : & 0 & 1 & 0 & 1 & 1 \\
 25 & : & 1 & 1 & 0 & 0 & 1 \\
 \hline
 11 \oplus 25 & : & 1 & 0 & 0 & 1 & 0
 \end{array}
 \end{array}$$

At 4th bit both are diff

At 1st bit both are diff

$$\underline{a \oplus b = 0} \quad \underline{a == b}$$

Given int Integer N, find least significant set bit position.

$$\begin{array}{r}
 \begin{array}{cccccc}
 & 5 & 4 & 3 & 2 & 1 & 0 \\
 N = 44 & : & 1 & 0 & 1 & 1 & 0 & 0
 \end{array}
 \end{array}$$

$$i = 0; i \leq 30; i++ \{$$

if (checkBit(N, i)) {

ans = i; break

}

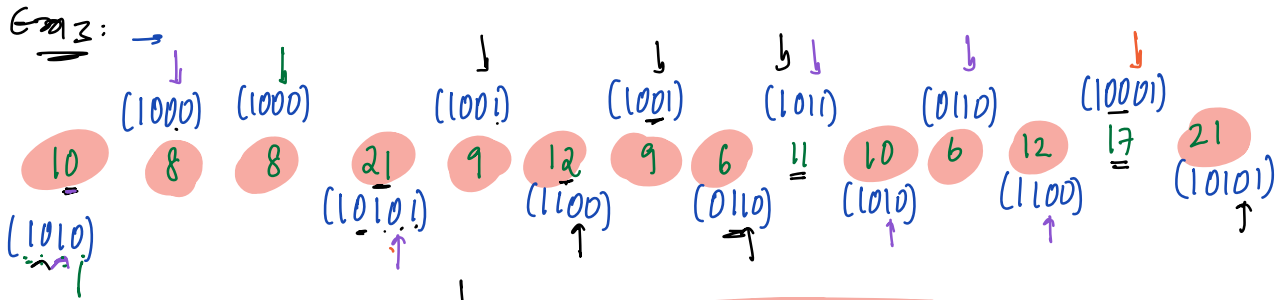
}

Given N array elements, every element repeats twice except 2 [unique elements], find The 2 unique elements

output:

$$\underline{Ex1}: \quad 3 \quad 4 \quad 5 \quad 4 \quad 7 \quad 5 \quad : \quad 3 \quad 7$$

$$\underline{Ex2}: \quad 4 \quad 9 \quad 9 \quad 8 \quad : \quad 4 \quad 8$$



Step 1: nor of all elements

→ nor of 2 unique elements

$$\begin{array}{cccccc} & 4 & 3 & 2 & 1 & 0 \\ \rightarrow 17 : & 1 & 0 & 0 & 1 & . \\ 11 : & 0 & 1 & 0 & 1 & 1 \end{array}$$

$$17 \oplus 11 : \underline{1} \underline{1} \underline{0} \underline{0} \underline{1} \underline{0}$$

At bit pos = 1, both 17 & 11 are different

Can we split on 0th bit: No

1st bit: YES

2nd bit: NO

3rd bit: YES

4th bit: YES

let's split array

1st bit set

$$\left\{ \begin{array}{l} 10, 6, 11, 10 \\ 6, \end{array} \right\}$$

nor of all elements

11

1st bit unset

unset

$$\left\{ \begin{array}{l} 8, 8, 21, 9, 12 \\ 9, 12, 17, 21 \end{array} \right\}$$

nor of all elements

17

let's split the array

3rd bit set

$$\left\{ \begin{array}{l} 10, 8, 8, 9, 12 \\ 9, 11, 10, 12 \end{array} \right\}$$

11

3rd bit unset

$$\left\{ \begin{array}{l} 21, 6, 6, 17, 21 \\ 17 \end{array} \right\}$$

Pseudo Code

1) val = xor of all Elements

2) pos = Get any set pos in
of val

3) Split based whether a
number is set or not
at posth

nn = 17 ^ 9

	4	3	2	1	0
17 :	1	0	0	0	1
9 :	0	1	0	0	1
	1	1	0	0	0

17

9

val = 0

for (i = 0; i < N; i++) {

 val = val ^ ar[i]

}

pos = 0

for (i = 0; i < 30; i++) {

 if (checkBit (val, i)) {

pos = i; break;

 }

}

set = 0, unset = 0

for (i = 0; i < N; i++) {

 if (checkBit (ar[i], pos)) {

set = set ^ ar[i]

 }

 else {

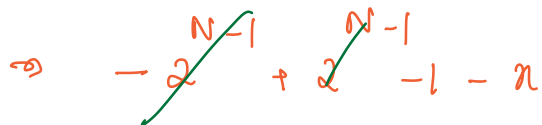
unset = unset ^ ar[i]

 }

}

print (set, unset)

$n \geq 0$ N bit {Doubt}

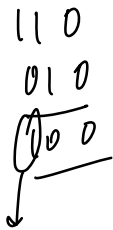
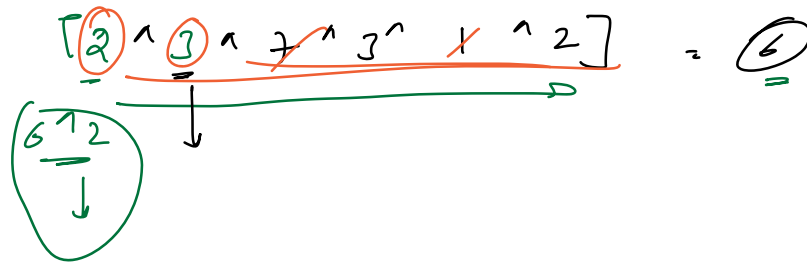


$$\sim x = -1 - x$$

$$-n = \sim n \uparrow$$

$$\sum_{i=1}^n x_i = n+1$$

Ex:



→ $p = 0; p < N; p_{++} \{$

$j = i; j < N; j_{++} \{$

$\text{if } (a[p] \wedge a[j] == 0) \{$

$\{$

$\}$

